

INSPACE-CAM-2026 Collision Avoidance Filing

Filing Reference ID:	7
Primary Spacecraft:	ISS (ZARYA) (NORAD 25544)
Secondary Spacecraft:	NORAD 900
Operator of Record:	OrbitGuard Swarm Agent Orchestrator
Time of Closest Approach (TCA):	2026-08-19 11:02:54.787509
Filing Status:	Filed
Submission Timestamp:	2026-08-20 01:40:10.086701

Regulatory Briefing & Maneuver Justification

****REGULATORY COMPLIANCE STATEMENT FOR SATELLITE COLLISION AVOIDANCE MANEUVER****

****Introduction****

This regulatory compliance statement is issued in accordance with applicable regulations and guidelines to inform stakeholders of the necessary measures taken to mitigate the risk of collision between two satellites, ISS (ZARYA) and CALSPHERE 1.

****Hazard Mitigation****

A collision between ISS (ZARYA) and CALSPHERE 1 poses a significant risk to both satellites and their respective operational capabilities. The primary objective of this maneuver is to prevent a potential collision, thereby ensuring the continued safe operation of both satellites and minimizing the risk of damage or destruction.

****Target Safety Window****

The target safety window for this maneuver has been established based on the predicted trajectory of both satellites. The Time of Closest Approach (TCA) is scheduled to occur on 2026-08-19T11:02:54.787509. The maneuver will be executed to ensure a safe separation of at least 0.35 km from the predicted TCA, thereby mitigating the risk of collision.

****Compliance with IADC Space Debris Guidelines****

This maneuver complies with the guidelines outlined in the International Designated Coordinate System for Space Operations (IADC) and the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) guidelines for space debris mitigation. Specifically, this maneuver adheres to the following guidelines:

* The maneuver will be executed within

Astrometric State Vectors & Covariance Matrices (TCA Epoch)

Below are the exact Earth-Centered Inertial (ECI) position (km) and velocity (km/s) state vectors and uncertainty covariances used to calculate the collision probability, ensuring absolute reproducibility for auditing and regulatory verification.

Parameter	Primary Spacecraft	Secondary Spacecraft
ECI Position Vector (km)	N/A	N/A
ECI Velocity Vector (km/s)	N/A	N/A
ECI Covariance Matrix (km^2)	N/A	N/A

Authorization & Sign-off

Authorized Operator Signature

Date